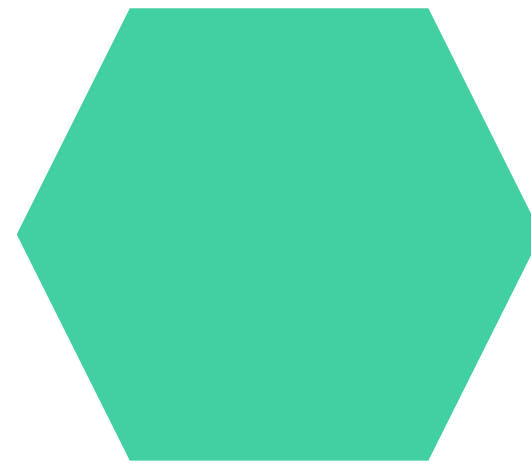
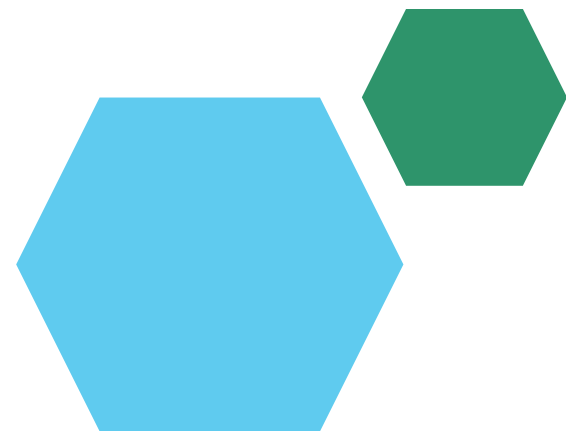
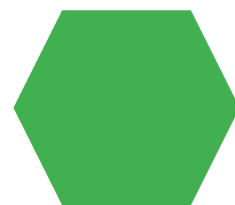




PROJECT TITLE- Seasonal Agriculture Performance Analysis

**[Student Name - Sayantanee Saha
College Name - Rajganj College]**

AICTE ID: (STU6a6c1794006131785468820)



PROBLEM STATEMENT

- Agricultural activities are influenced by seasonal variations in environmental conditions.
- Farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another.
- Raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions.
- The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.

Project Description

Seasonal Agriculture Performance Analysis is an end-to-end data analytics and machine learning project evaluating crop performance across Kharif, Rabi, and Zaid seasons.

- Data Preprocessing & Feature Engineering: Cleaned raw data, handled missing values, and created key metrics like Profit/Revenue per Hectare and Total NPK usage.
- Exploratory Data Analysis (EDA): Analyzed seasonal trends, regional variations, and resource usage using Pandas, NumPy, Matplotlib, and Seaborn.
- Statistical Validation: Applied ANOVA, Kruskal-Wallis, and correlation analysis via SciPy to validate relationships between environmental factors and yield.
- Machine Learning & Insights: Built a Random Forest Regression model (evaluated using R^2 and MAE) to predict yield and provide data-driven farming insights.

WHO ARE THE END USERS?

- **Farmers:** Use insights for crop selection, seasonal planning, yield improvement, and efficient resource utilisation
- **Government & Agriculture Departments:** Support agricultural policies, schemes, resource distribution, and sustainable farming initiatives.
- **Researchers:** Study seasonal variations, crop performance, environmental factors, and agricultural trends.
- **Data Analyst:** Use the dataset and analytical results to identify patterns, relationships, and performance indicators.
- **Agribusinesses:** Identify profitable crop-season combinations and improve production and investment decisions.
- **Irrigation Manager:** Analyze water usage and efficiency to support better irrigation planning.
- **Environmental & Sustainability Teams:** Evaluate resource consumption and environmental conditions to encourage sustainable agricultural practices

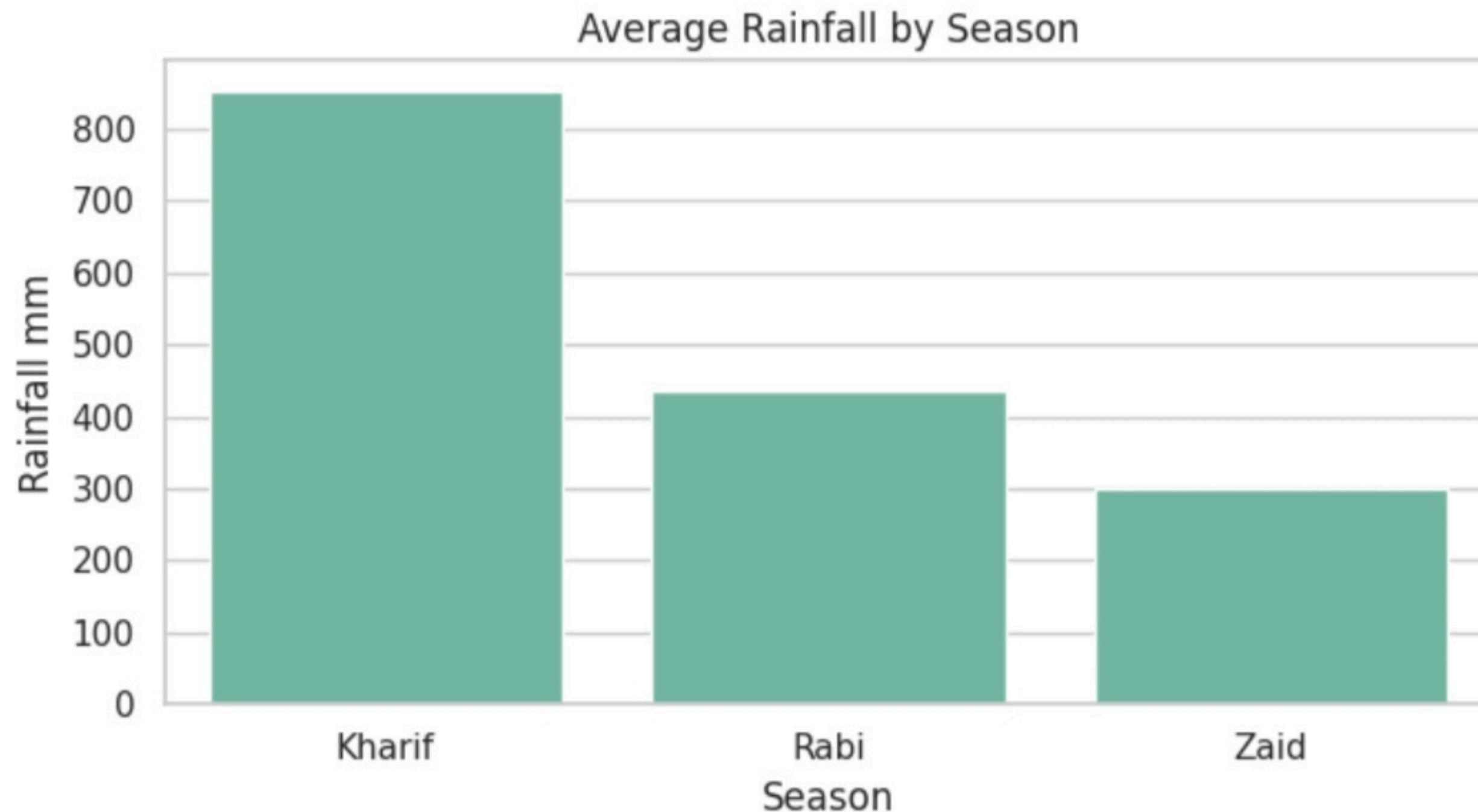
Technology Used

- **Python:** Core programming language for analysis, stats, and ML.
- **Google Colab/Jupyter Notebook:** Development environment for execution and documentation.
- **Pandas:** Data loading, cleaning, transformation, and aggregation.
- **Numpy:** Numerical operations and feature calculations.
- **Matplotlib:** Basic charts and data visualizations.
- **Seaborn:** Advanced statistical charts (box plots, heatmaps).
- **Scipy:** Statistical hypothesis testing (ANOVA, Kruskal-Wallis, Mann-Whitney).
- **Scikit-learn:** Model building and evaluation for yield prediction.
- **Random Forest:** Feature importance analysis and yield estimation
- **Statistical Analysis:** Seasonal validation and significance testing.
- **Correlation Analysis:** Environmental and resource relationship tracking.
- **Feature Engineering:** Key metric creation (Profit/Revenue/Water per hectare, Total NPK).
- **CSV/Excel:** Source dataset storage and processing format.



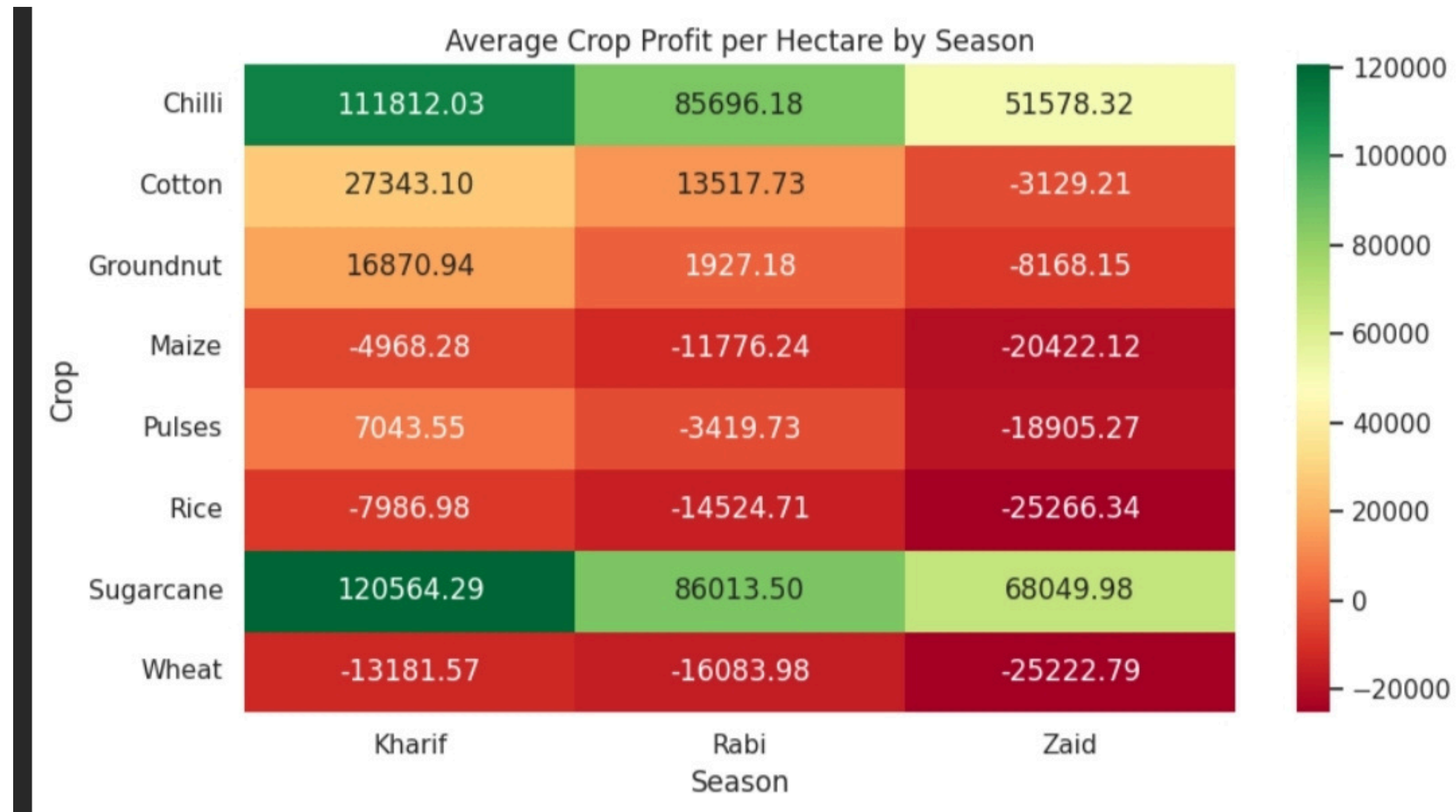
RESULTS

- Kharif receives peak rainfall (>800mm), aiding rainfed crops.
- Zaid gets the least rainfall (approx 300mm), increasing irrigation dependency



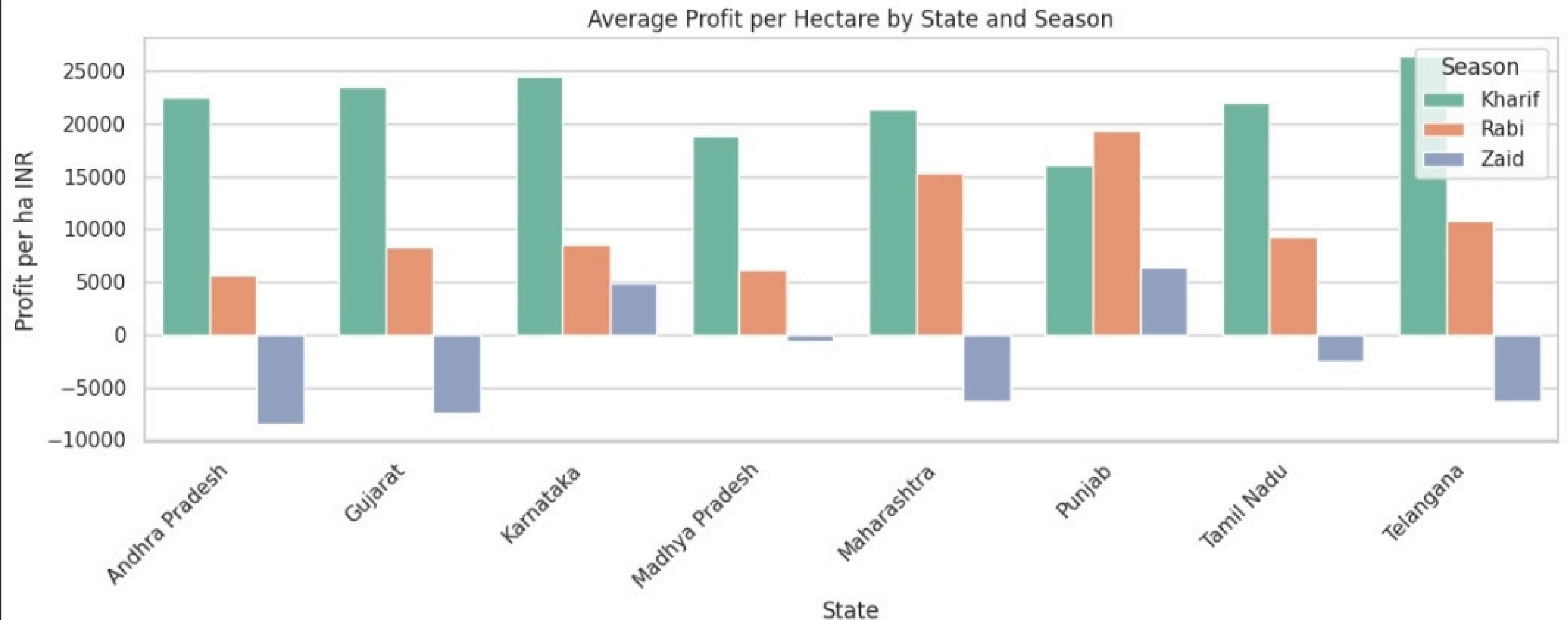
RESULTS

- Sugarcane and Chilli generate the highest profits, especially in Kharif.
- Wheat, Rice, and Maize record negative profits across all seasons.



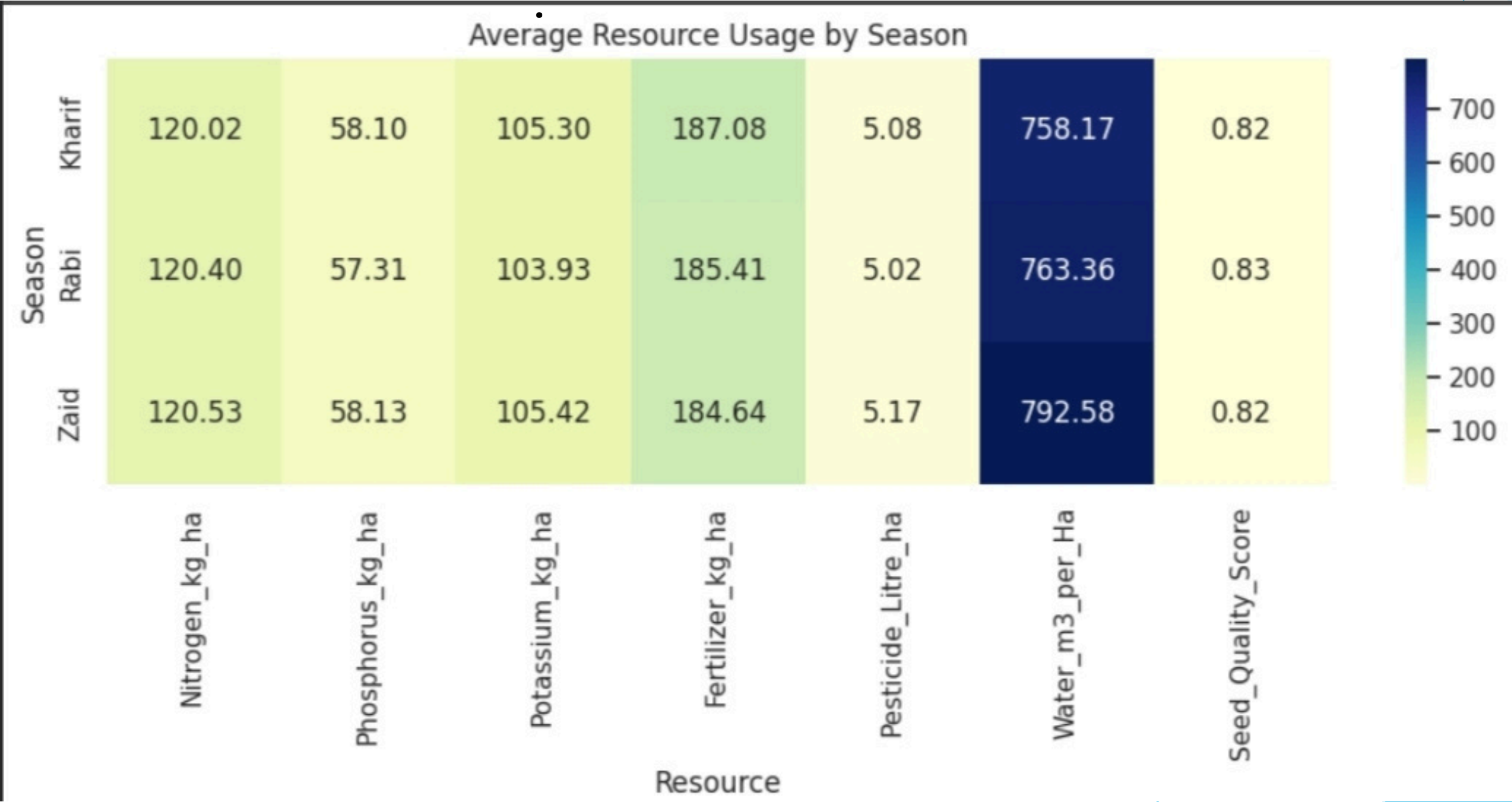
RESULTS

- Kharif brings the highest profits for most states, led by Telangana.
- Zaid shows negative profits overall, except in Punjab and Karnataka



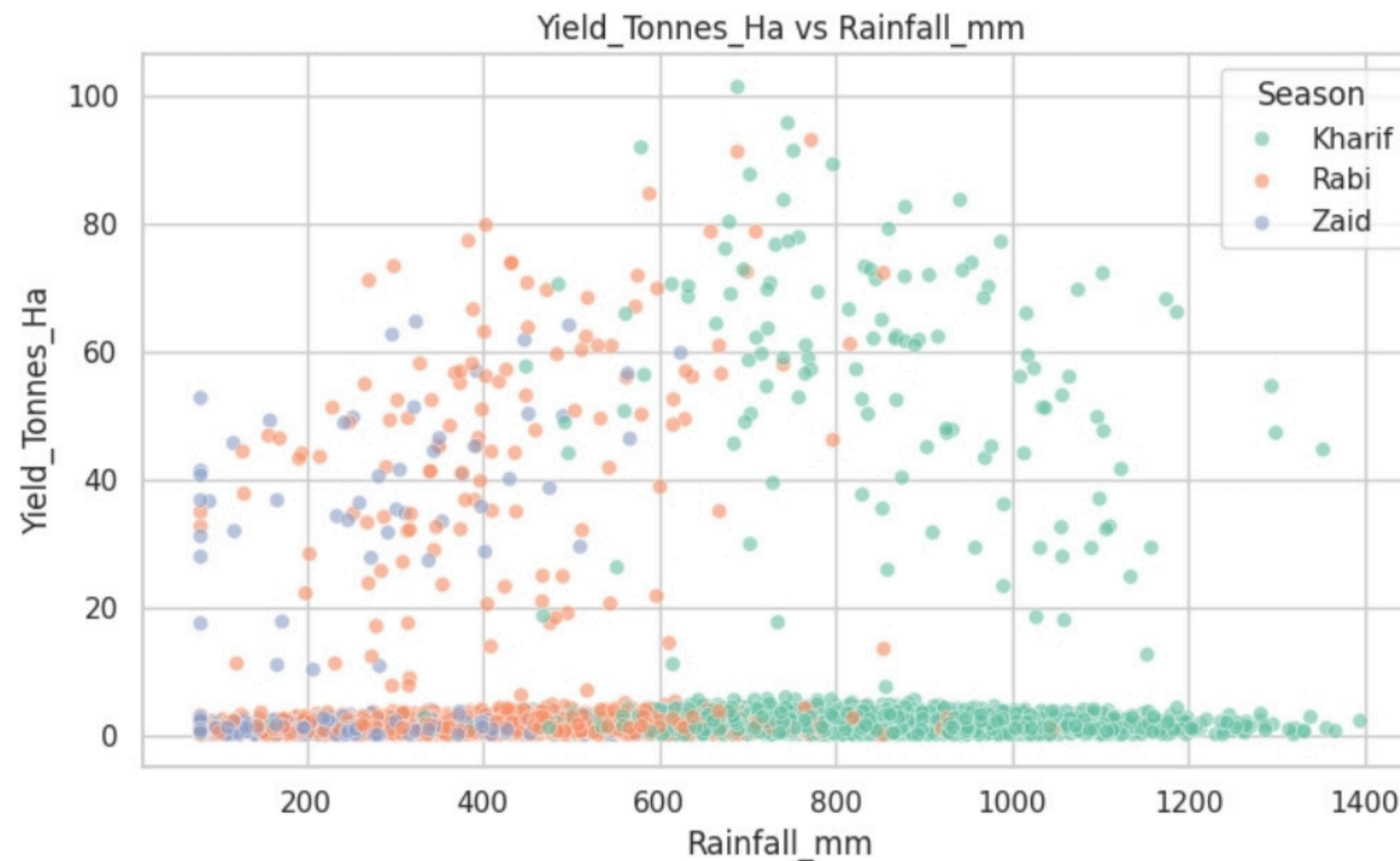
RESULTS

- Fertilizer and pesticide usage remain constant across seasons.
- Water consumption is highest in Zaid (792.58m³/Ha).



RESULTS

- Maximum crop yield occurs in Kharif with rainfall between 600--1000mm.
- Low rainfall in Rabi and Zaid results in lower overall yields.



Future scope

- 
- Seasonal Analysis: Compare agricultural performance across Kharif, Rabi, and Zaid seasons.
 - Crop Profitability: Identify the most profitable and loss-making crop-season combinations.
 - FieldPrediction: Predict crop yield using environmental, soil, and resource-related factors.
 - Crop Recommendation: Suggest suitable crops based on season, expected yield, profitability, and resource availability.
 - Water Management: Analyze water consumption and improve water-use efficiency.
 - Risk Analysis: Evaluate disease and pest risks across crops and seasons.
 - Resource Optimization: Improve the use of water, fertilizer, pesticide, and other agricultural inputs.
 - Regional Analysis: Compare crop performance and profitability across different states.
 - Interactive Dashboard: Present key insights through charts, rankings, heatmaps, and performance indicators.
 - Future Scope: Extend the system with real-time weather data, market prices, advanced ML models, and personalized farmer recommendations.
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GitHub Link

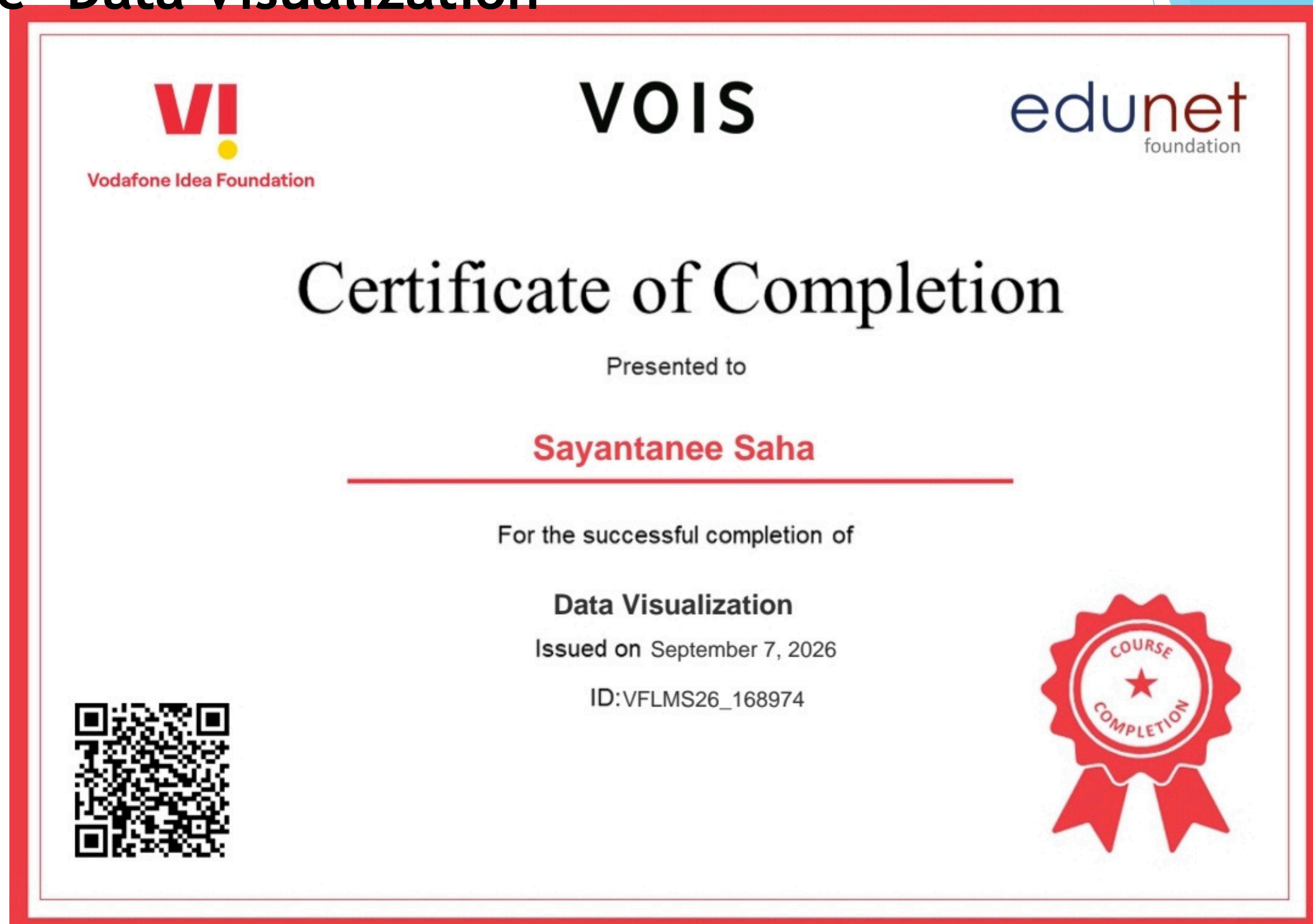


<https://github.com/Sayantane-cs/Seasonal-Agriculture-Performance-Analysis->



VOIS Course completion certificate -

Course name- Data Visualization



THANK YOU

THANK YOU FOR THIS WONDERFUL OPPORTUNITY

